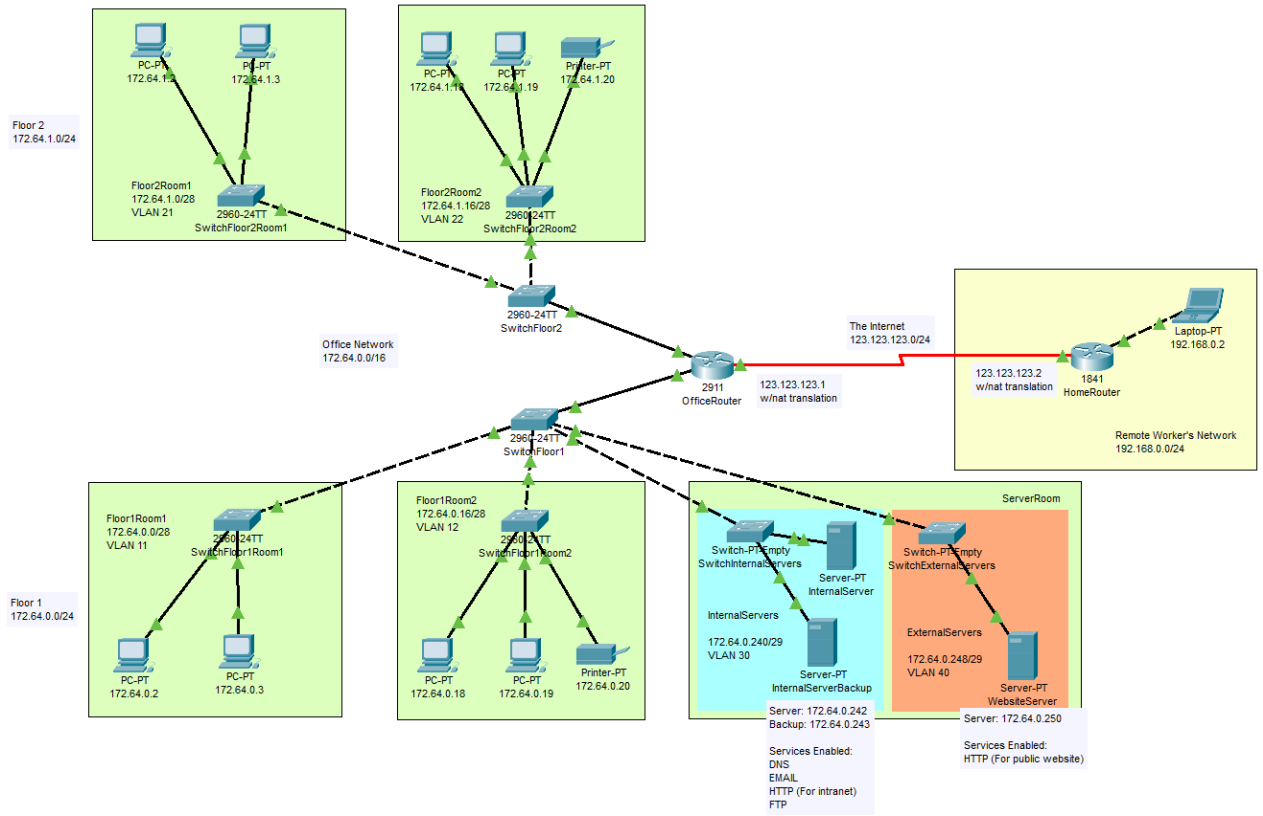
Network Design

a)

This image shows the packet tracer design for my network.



Assumptions:

- The real office will contain 150 computers, using this information:
    - Each floor will have more rooms (max 16 per floor with current subnetting).
    - Each office room will have more devices (max 13 per room with current subnetting).

Important points:

- I included a public web server for hosting the Teclyn website where the company can advertise their services, clients will be able to learn about and contact the company for technology solutions.
- A private internal server hosts an intranet that could be used for employee services like viewing payslips and making HR requests, or as a tool to assist in project tracking and collaboration. It also hosts employee files and an email system, ensuring a secure centralised location for Teclyn's data.
- Printing services are included on each floor so staff can print documents over the network.
- Remote workers can connect to the network over the internet to enable working from home (not currently working).

b)

These spreadsheet images show the subnetting scheme of my office network.

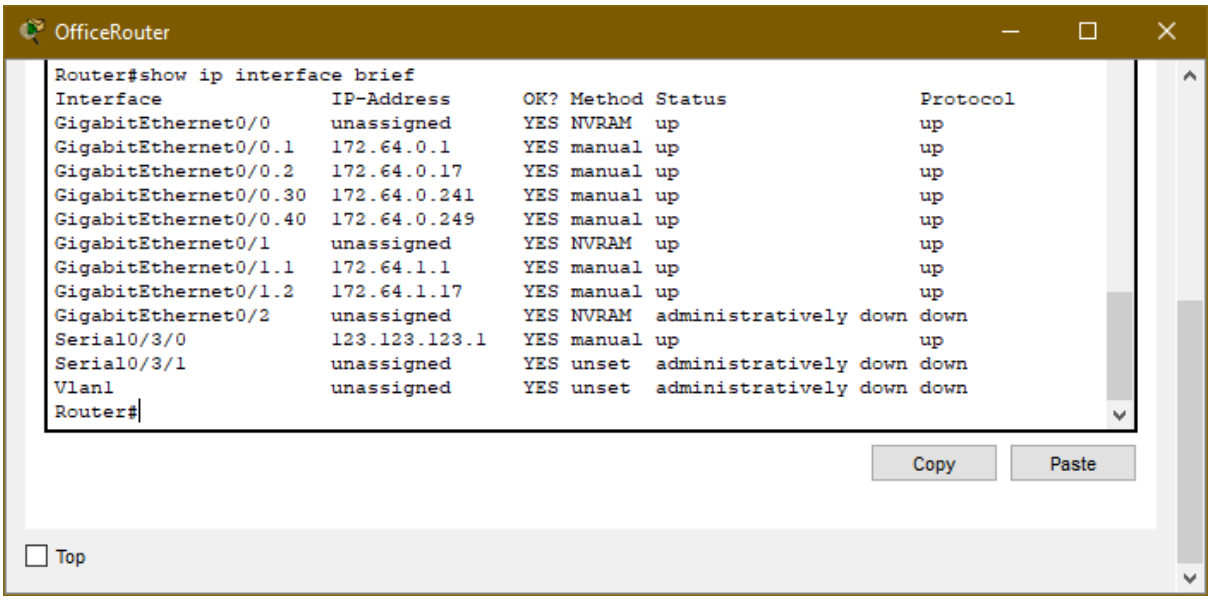
Office Network							
Floor 1				Floor 2			
172.64.0.0	0	network	Floor1Room1 VLAN 11 255.255.255.240	172.64.1.0	0	network	Floor2Room1 VLAN 21 255.255.255.240
172.64.0.1	1	router		172.64.1.1	1	router	
172.64.0.2	2	pc		172.64.1.2	2	pc	
172.64.0.3	3	pc		172.64.1.3	3	pc	
172.64.0.4	4			172.64.1.4	4		
172.64.0.5	5			172.64.1.5	5		
172.64.0.6	6			172.64.1.6	6		
172.64.0.7	7			172.64.1.7	7		
172.64.0.8	8			172.64.1.8	8		
172.64.0.9	9			172.64.1.9	9		
172.64.0.10	10			172.64.1.10	10		
172.64.0.11	11			172.64.1.11	11		
172.64.0.12	12			172.64.1.12	12		
172.64.0.13	13			172.64.1.13	13		
172.64.0.14	14			172.64.1.14	14		
172.64.0.15	15	broadcast		172.64.1.15	15	broadcast	
172.64.0.16	0	network	Floor1Room2 VLAN 12 255.255.255.240	172.64.1.16	0	network	Floor2Room2 VLAN 22 255.255.255.240
172.64.0.17	1	router		172.64.1.17	1	router	
172.64.0.18	2	pc		172.64.1.18	2	pc	
172.64.0.19	3	pc		172.64.1.19	3	pc	
172.64.0.20	4	printer		172.64.1.20	4		
172.64.0.21	5			172.64.1.21	5		
172.64.0.22	6			172.64.1.22	6		
172.64.0.23	7			172.64.1.23	7		
172.64.0.24	8			172.64.1.24	8		
172.64.0.25	9			172.64.1.25	9		
172.64.0.26	10			172.64.1.26	10		
172.64.0.27	11			172.64.1.27	11		
172.64.0.28	12			172.64.1.28	12		
172.64.0.29	13			172.64.1.29	13		
172.64.0.30	14			172.64.1.30	14		
172.64.0.31	15	broadcast		172.64.1.31	15	broadcast	
172.64.0.240	0	network	InternalServers VLAN 30 255.255.255.248	172.64.1.240	0	network	
172.64.0.241	1	router		172.64.1.241	1		
172.64.0.242	2	internalserver		172.64.1.242	2		
172.64.0.243	3	internalserverbackup		172.64.1.243	3		
172.64.0.244	4			172.64.1.244	4		
172.64.0.245	5			172.64.1.245	5		
172.64.0.246	6			172.64.1.246	6		
172.64.0.247	7	broadcast		172.64.1.247	7		
172.64.0.248	0	network	ExternalServers VLAN 40 255.255.255.248	172.64.1.248	8		
172.64.0.249	1	router		172.64.1.249	9		
172.64.0.250	2	websiteserver		172.64.1.250	10		
172.64.0.251	3			172.64.1.251	11		
172.64.0.252	4			172.64.1.252	12		
172.64.0.253	5			172.64.1.253	13		
172.64.0.254	6			172.64.1.254	14		
172.64.0.255	7	broadcast		172.64.1.255	15	broadcast	

IPs not included in the images are unassigned.

Coloured sections indicate assigned subnets.

Important points:

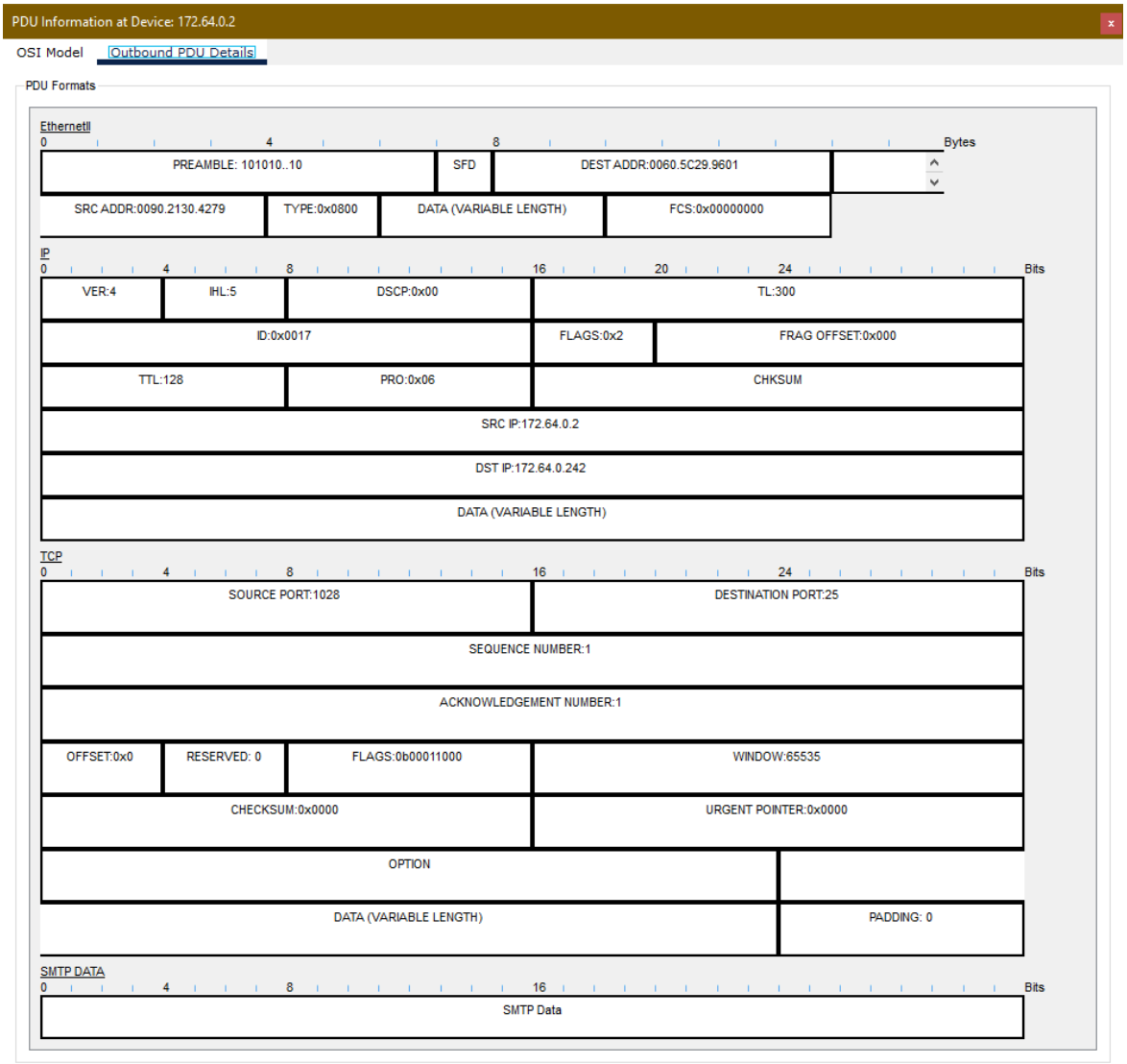
- The first and second floors of the office are logically separated into subnets 172.64.0.0/24 and 172.64.1.0/24 respectively (not explicitly via interfaces as VLANs/subnets are used within each range).
- Although the network is based on 172.64.0.0/16, immediately splitting it into two /24 subnets is not wasteful given the company's size and needs. Each /24 subnet provides 254 usable IP addresses, which is ample for the current requirements of each floor while leaving room for expansion. The /16 base network ensures plenty of address space is available for future subnetting if the company grows, and splitting into /24 subnets keeps the network simple, manageable, and scalable without significant waste.
- The office rooms are assigned /28 subnets (255.255.255.240 subnet mask). These subnets have corresponding VLANs, reducing unnecessary network traffic, enhancing security by isolating each room, and simplifying network management. According to my assumptions that Teclyn organises their office with relatively small office rooms, the 13 usable IPs per room is appropriate (+1 IP for router subinterface).
- I have used variable subnetting on floor 1 to implement the server room. The server room is divided into two /29 subnets and two corresponding VLANS. One subnet is dedicated to private-facing servers and the other is for public-facing servers. This is for security and performance reasons discussed later. These subnets allow 5 servers each. This is ideal for future expansion whilst not taking too many IP's from the floor's subnet.
- These are the subinterfaces used for the subnets:



Networking Services and Protocols

a)

To demonstrate the process of networking services and protocols, I'll use the following image, a snapshot of an SMTP packet sent by a Teclyn employee's PC at 172.64.0.2, while explaining the process of sending an email on the network:



- The employee clicks 'send' on the email client. The email client invokes the SMTP protocol, a Layer 7 (Application Layer) protocol that handles formatting and presentation of the email's content.
- In Layer 6, the Presentation Layer, the email data is compressed to reduce its size and encrypted to ensure privacy and security during transmission.
- Layer 5, the Session Layer, authenticates the device and opens a session, ready to manage dialogue between the two devices.
- Layer 4, the Transport Layer, is the first layer to use encapsulation, where headers are added to the data to facilitate its transmission. The data is split into smaller segments known as PDUs (Protocol Data Units), each assigned a sequence number for reordering and restoration at the destination. The sequence number of the illustrated packet is '1'. Port numbers are also assigned in the header. In this case,

the source port number is dynamically assigned (1028), and the destination port number is 25 (SMTP).
- In Layer 3, the Network Layer, a source IP address (172.64.0.2) and a destination IP address (172.64.0.242) are added by the IPv4 protocol, creating packets. A protocol flag (0x06 for TCP) is set to indicate the transport protocol in use.
- In Layer 2, the Data Link Layer, the Ethernet protocol encapsulates the data into a frame. The next-hop MAC address is added; as the email server is in a different subnet, the MAC address of the router (0060.5C29.9601) is used first. This is what the illustrated packet shows as it has just left the PC. After reaching the router, the next hop address will be the server's MAC address (0001.64CB.1904). A CRC checksum is appended to verify data integrity.
- At Layer 1, the Physical Layer, the data is converted to electrical signals for transmission over Teclyn's copper cables.

Unlike SMTP, to ping another device on the network, the protocol used is ICMP, which operates at Layer 3 (Network Layer) without requiring any Transport or Application Layer encapsulation. It's a 'connectionless protocol', meaning the source device never opens a formal connection with the destination device prior to sending.

For uploading files to Teclyn's internal server, FTP is used at the Application Layer. Unlike SMTP, FTP establishes two sessions at the Transport Layer: one for commands (port 21) and one for data transfer (port 20).

If the protocol being used is interacting with a device on the same network/subnet, e.g. 172.64.0.2 to 172.64.0.3, data can be transmitted over the switch using the MAC address rather than using the router and the IP address. This ensures faster and more efficient communication.

Discussed protocols are implemented and working in my design, as shown in my video.

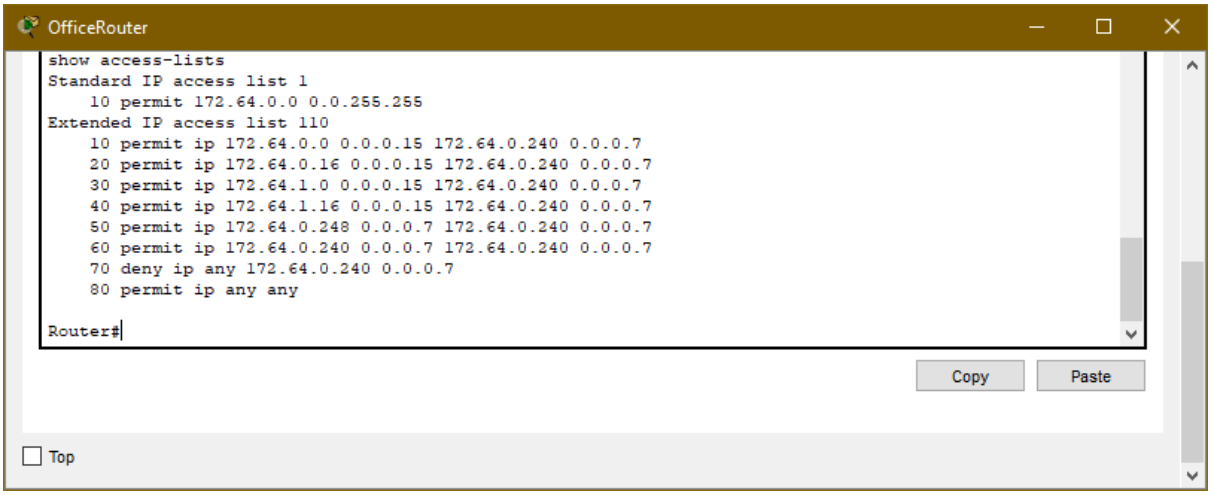
Network Security

a)

My design aims to uphold the principles of information security - confidentiality, integrity, and availability.

My design employs a strong password policy for the office router and switches (implemented), with 2FA (not implemented) to further reduce risks of unauthorised access. Employees are reminded to keep passwords private and avoid phishing attempts. 2FA increases confidentiality, and securing networking hardware reduces the risk of faults in availability from cyberattacks or misconfigurations caused by inexperienced staff.

To control and monitor network traffic, an ACL is implemented on the router. This blocks unauthorised traffic at the network layer, only allowing office subnets access to the internal server subnet. For example, if a client or malicious actor uses the network outside of an employee subnet, they are completely blocked from internal servers. Logging attempted connections (unavailable in Packet Tracer) provides valuable insights into attempted breaches, enhancing accountability and informing security measures. Lowering chances of tampering to the internal server, confidentiality and data integrity are increased.



A RADIUS server (not implemented) centralises authentication, authorisation, and accounting. It assigns unique login credentials to employees, tied to specific permissions and access controls. The employee intranet's 2FA login system (not implemented) further prevents unauthorised access to company tools and information.

Office rooms are assigned separate VLANs for more granular traffic control. This isolates compromised devices, VLAN isolation limits the attacker's ability to access devices in other rooms. Internal and external servers are also in separate VLANs to protect sensitive resources from public-facing systems, helping to improve confidentiality.

Availability on the network is increased with a backup internal server. In case of primary server failure due to a DDoS attack, malware, or hardware issues, Teclyn can quickly switch to the backup server, significantly reducing downtime and permanent data loss. Other physical features will be implemented too, such as physically securing the server room and locking networking racks.

In the design, antimalware services are installed on devices across the network (not implemented). Malware such as trojans, spyware, or keyloggers could be used to steal company data, threatening confidentiality, or corrupt files, threatening both availability and integrity. This software as well as the OS of all of Teclyn's PCs will be kept up to date with the latest security patches to provide increased protection.

The design uses a VPN (not implemented) to securely connect remote workers to the office network. The VPN encrypts communication, ensuring data confidentiality and protecting sensitive resources from interception. Additionally, using SSH and SCP (not implemented) alongside the VPN enables secure device management and file transfers, safeguarding the integrity of transmitted data. The use of the systems in tandem provides remote workers with flexibility in using the office network whilst ensuring high security for important actions like uploading files to the internal server. As Casto (2019) explains, a VPN server facilitates remote access, while enabling SSH servers on devices enhances security and management. On top of this, use of NAC (not implemented) ensures that devices are fit to connect to the Teclyn network, for example verifying the previously mentioned security and OS updates have been installed.

Network Performance

a)

Congestion on a network occurs when an interface becomes overloaded with traffic, leading to delays and reduced performance. My network uses a dedicated switch for each floor, splitting the traffic load between two switches instead of relying on one. This design reduces the risk of congestion caused by aggregation, where traffic from many sources (the room VLANs) is routed over one interface (to the router). I also ensured the use of high-speed (Gb) cables from the switches to the router to provide sufficient throughput for floor traffic, preventing the switches from becoming overwhelmed.

The room VLANs reduce congestion by segmenting traffic and minimising broadcast traffic across the network. Isolating public and private servers using VLANs then further minimises interference between customer web traffic and internal operations.

The network has a Class-Based Weighted Fair Queuing QoS policy (not implemented) to prioritise critical traffic like video and voice calls, essential for client communication. This policy reduces jitter and packet loss, ensuring professional interactions. It uses the DiffServ QoS model, which allows for scalable traffic prioritisation through class types, and ensures flexibility as the network grows. The network uses a high-bandwidth ISP (not implemented) to avoid the external connection becoming a bottleneck, especially important for the web server and client connections that require minimal response-times.

After implementing the network, I would conduct 3-week baselining to evaluate normal network performance. During this time I'd measure key metrics like response-time, latency, round-trip time, and packet loss. These will help me identify areas of congestion and underutilisation, enabling reconfiguration of hardware settings or my VLAN/subnet scheme.

b)

The design features a backup internal server for redundancy, vital for maximising uptime in the event of a hardware issue.

To help prevent interface or cable faults affecting availability, the network features EtherChannel link aggregation (not implemented) between all switches, combining two physical copper cables into one logical link, increasing bandwidth and ensuring redundancy if one link fails. It also features N+1 redundancy with STP (not implemented), adding fault tolerance by pairing each switch with a backup and preventing Layer 2 loops by selecting most efficient paths whilst reserving others for redundancy.

As Casey (2024) mentions, maintenance like updating firmware and inspecting hardware is key to minimising downtime. The modular design with clear room segmentation (VLANs/subnets) should greatly simplify this maintenance and configuration. The use of a switch for each room also means that even if the backup somehow fails, much of the network remains operational due to isolated room-level configurations.

Lamberti (2023) highlights some of the key metrics for measuring dependability:

- Availability: The percentage of time a network is operational, the design aims for an availability of five nines (99.999%).

- Mean Time Between Failures (MTBF): The average time between network errors, this should be high to reflect Teclyn's reliability standards.
- Mean Time to Repair (MTTR): The average time needed to resolve errors. For example, if the internal server experiences a malfunction, the goal is to restore functionality within 10 minutes using the backup server.
- Redundancy Metrics: Evaluating network redundancy, such as my design's N+1 redundancy (N components with 1 backup) (not implemented).

References

Casto, D. 2019. Using SSH and VPN to Access Your Home Network | Available at: https://homenetworkguy.com/tech/ssh-vs-vpn/#:~:text=via%20firewall%20rules.-,Using%20a%20VPN%20and%20SSH%20Together,facing%20services%20on%20your%20network

Casey, K. 2024. How to Improve Network Reliability | Available at: https://thectoclub.com/uncategorized/network-reliability/

Lamberti, A. 2023. Enhancing Network Reliability: How to Measure, Test & Improve It | Available at: https://obkio.com/blog/how-to-measure-improve-network-reliability/#measuring-network-reliability-understanding-your-network